

Chain Forward

Risk Assessment & Management Framework for Value Chain MSME Lending

Forward fintech

Prepared by: Faith Wanjiru Kirigi

Executive Summary

Financial viability: Base case generates **\$1.2M annual contribution** after expected credit losses, but does not recover the \$4.5M investment within 3 years.

Credit risk: Default risk is highly sensitive to portfolio performance; the modeled 6% default rate produces a **45-month payback**.

Data science: Behavioral transaction features provide useful risk discrimination, with the logistic regression achieving **0.837 ROC-AUC**

Recommendation: Proceed cautiously with a controlled pilot, risk-based underwriting and pricing, and portfolio-level monitoring.

Business Challenge & Risk Framework

Credit Risk Thin-file MSMEs: score via behavioral & transaction data, not financials.	Value Chain Risk Score the chain, not just the borrower — a supplier's risk depends on the retailers/distributors it transacts with.
Behavioral Risk Transaction patterns and volatility signal repayment capacity.	Liquidity Risk Cohort-level cash flow stress testing given 7–90 day tenor and high repayment frequency.
Concentration risk Cap exposure per value chain / anchor buyer; one failure can cascade through connected borrowers.	Alternative data integration Blend mobile money, utility, and geolocation signals into a single score where formal data is sparse.

Risk Hypotheses & Data Strategy

Hypothesis	Data	Method
Higher cash-flow volatility increases default risk	Transaction history	Logistic regression
Frequent/consistent transactions indicate stronger repayment capacity	Transaction history	Clustering + classification
Repayment behavior predicts future default	Loan/repayment data	Logistic regression
Certain MSME behavioral segments are inherently riskier	Transaction data	K-Means
Value-chain concentration can amplify portfolio losses	Customer/value-chain relationships	Network analysis

Financial risk model

Metric	Base Case
Annual contribution	\$1.20M
3-year contribution	\$3.60M
3-year ROI	-20%
Payback period	45 months

Base case does not achieve the 3-year (36-month) payback target — it lands at 45 months.

Scenario Analysis

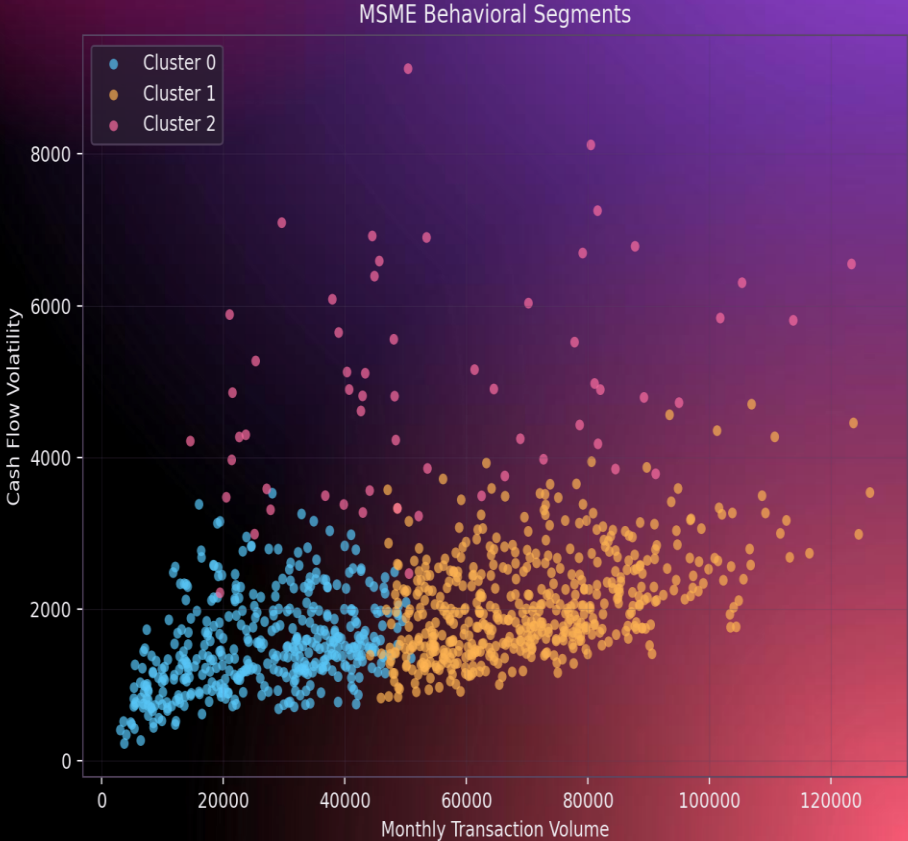
	Optimistic	Base	Pessimistic
Borrowers	22K	20K	18K
Default rate	4%	6%	10%
LGD	40%	50%	60%
Monthly contribution	\$162.8K	\$100K	\$18K
3-year ROI	30%	-20%	-86%
Payback	27.6 mo.	45 mo.	250 mo.

Portfolio profitability is highly sensitive to credit performance.

MSME Customer Segmentation

Segment	Frequ ency	Avg Transaction	Monthly Volume	Volatility
Cluster 0: Stable / Lower Activity	16.9	\$1,593	\$26.6K	\$1,511
Cluster 1: High Activity	38.9	\$1,829	\$70.5K	\$2,077
Cluster 2: High Value / Volatile	18.7	\$3,158	\$55.6K	\$4,915

Cluster 2 warrants enhanced monitoring because high transaction value is accompanied by substantially higher cash-flow volatility.

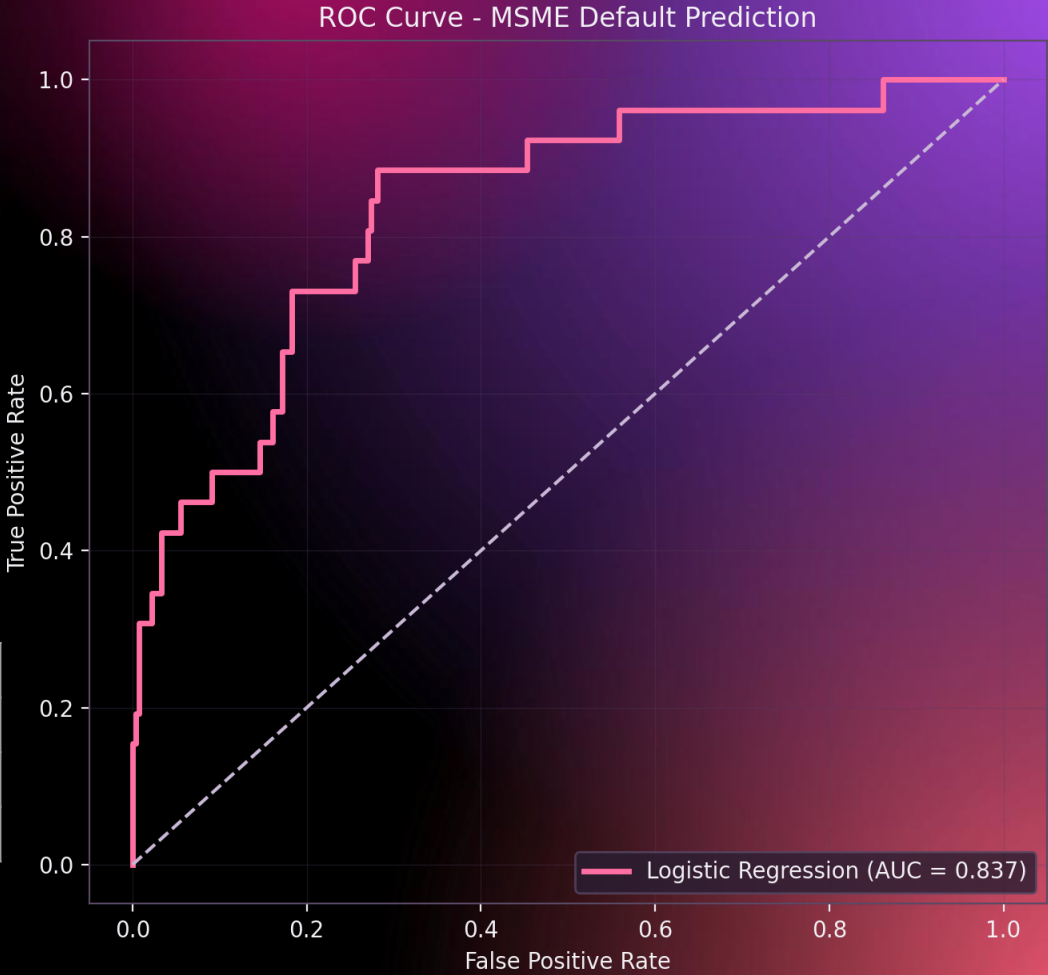


DEFAULT PREDICTION & RISK-BASED PRICING

Metric	Result
ROC-AUC	0.837
Precision	0.727
Recall	0.308
F1	0.432

Illustrative risk-based pricing

Segment	Share	Rate/mo
Cluster 0 – Stable	42%	\$8.00
Cluster 1 – Active	52%	\$9.50
Cluster 2 – Volatile	6%	\$16.00



EARLY WARNING & MONITORING FRAMEWORK

Model risk drivers

Feature	Effect on risk
Cash-flow volatility	↑ risk (strongest driver)
Transaction frequency	↓ risk (protective)
Repayment ratio	↓ risk (protective)
Monthly transaction volume	↑ risk (mild)
Avg. transaction amount	↓ risk (protective)

Cash-flow volatility is the strongest modeled risk signal, suggesting that changes in transaction stability should be monitored as an early warning indicator.

Early warning indicators

Behavioral

- Rising cash-flow volatility
- Declining transaction frequency
- Falling repayment ratio
- Unusual transaction patterns

Credit

- Missed/late repayments
- Increasing utilization
- Repeat borrowing frequency

Portfolio

- Default rate by segment
- Vintage/cohort default rate
- Value-chain concentration
- Geographic concentration

Close the payback gap: credit losses clear the 16% break-even default rate with room to spare, but the payback period still misses the 36-month target by **\$25K/month**. Closing it needs roughly a +16% yield lift (to \$9.25/borrower), a lower 29% LGD, or a 3.5% default rate, pricing and collections are more realistic levers than underwriting alone.

Tighten controls on Cluster 2: the High Value/Volatile segment (60 customers) carries a risk score **more than 3x** Cluster 0's, driven by cash-flow volatility despite higher transaction value, apply tighter exposure limits and closer monitoring here before scaling.

Support the model, don't rely on it alone: 0.837 ROC-AUC comes with only **0.308 recall** at the current threshold it misses most actual defaulters. Pair the score with manual review for borderline cases, and recalibrate once real repayment data replaces the synthetic labels.

Next steps: launch a controlled pilot with the early-warning framework (slide 9) live from day one, revisit the \$100 EAD and 50% LGD assumptions against real chain data within the first quarter, and gate any scale-up on the pilot clearing the 36-month payback target.